The Double-Slit Experiment on a Discrete Holographic Lattice: Interference, Decoherence, and the Emergence of Classical Probability

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Abstract

We show that exact unitary propagation of an electron wavepacket on a finite 4.8.8 lattice reproduces single-slit and double-slit diffraction patterns in close quantitative agreement with experiment, with no continuum approximation. When environmental coupling is introduced at the slit, interference is destroyed and Born rule statistics emerge from decoherence alone. Using the deterministic kinematics of the circlette lattice, we demonstrate that wavefunction collapse need not be postulated. The transition from quantum interference to classical probability distributions emerges naturally from unitary evolution through the deterministic tracing-out of environmental degrees of freedom. The lattice addresses the apparent measurement problem through finite bandwidth and orthogonal geometry, requiring no conscious observer. This is the third paper in a series developing the Holographic Circlette framework, but remains mathematically self-contained.

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1 Introduction

The double-slit experiment is the fundamental touchstone for wave-particle duality. From Thomas Young's original demonstration of light interference, to Richard Feynman's assertion that the electron double-slit contains the "only mystery" of the quantum world [1], it has served as the ultimate testbed for physical theories.

Despite its ubiquity, the exact analytical calculation of double-slit diffraction for realistic physical geometries remains essentially unsolved in closed form. Standard treatments rely on the Fraunhofer approximation. Addressing the near-field Fresnel regime requires evaluating the Kirchhoff-Fresnel integral, which rapidly becomes intractable for slits of finite width and depth. The deeper issue lies in the boundary conditions: as established by Sommerfeld's half-plane solution [2], the electromagnetic field equations in continuous space become singular at perfectly sharp, conducting edges—the so-called "problem of the screen."

While Feynman's path integral formulation is conceptually elegant, numerical evaluations face severe computational bottlenecks. Evaluating the uncountably infinite sum over all continuous paths through a physical slit geometry is notoriously difficult, as Monte Carlo path sampling approaches are inherently plagued by the numerical sign problem, where oscillating complex phases lead to exponential cancellation errors [3].

This paper delivers an exact computational solution by proposing that the physical universe does not execute continuum limits. By modelling spacetime fundamentally as a discrete lattice, the path integral ceases to be an uncountably infinite sum. A slit edge is not a mathematical singularity; it is simply a topological boundary where propagation rules terminate.

This is the third in a series of papers developing the Holographic Circlette lattice model [9, 10]. Parts I and II established the lattice structure and its derivation of the Standard Model. Here we show that the same lattice, with no additional assumptions, natively reproduces exact macroscopic wave optics and quantum decoherence.

2 The Circlette Lattice

We briefly review the kinematic substrate required for this simulation. Spacetime is modelled as a discrete 2D 4.8.8 (truncated square) tiling. The state of a particle is described by a 4-component complex spinor residing on the octagonal sites, representing the physical combinations of spatial routing (Left/Right, Up/Down).

Because we are simulating a stable electron, the weak-force CNOT gate remains dormant. The quantum walk is driven entirely by spatial shifting and chirality mixing. The local unitary "coin" operator that applies this mixing is:

$$U(m) = \cos(m)I - i\sin(m)\sigma_x \tag{1}$$

where m is a dimensionless mass parameter defining the chirality toggling angle (in radians) per Planck tick, and σ_x couples counter-propagating spatial channels. This continuous unitary toggling imparts inertia and subluminal group velocity, natively generating dispersive *Zitterbewegung* [9].

3 Wavepacket Propagation on the Lattice

To avoid the exponential scaling and numerical sign problems inherent in Monte Carlo path sampling, we compute the direct unitary evolution of the entire lattice simultaneously via explicit finite-difference steps. Because quantum evolution is strictly linear, propagating the full amplitude field for T steps automatically and exactly computes the coherent sum over all lattice paths of length T, without requiring explicit path enumeration.

The wavepacket is initialised as a finite Gaussian distribution. Evolution consists of two steps per lattice tick: shifting amplitudes to adjacent sites, and applying the local coin operator $U(m)$. To prevent unphysical reflections, the extreme boundaries of the lattice act as a Perfectly Matched Layer (PML), applying a $\sin^2$ attenuation envelope. The physical slits are defined simply by a boolean mask.

4 Single-Slit Diffraction

Initial validation of the discrete kinematics was performed using a single slit. As the wavepacket squeezes through the topological aperture, the discrete lattice natively bridges the near-field and far-field regimes, seamlessly demonstrating the Fresnel-to-Fraunhofer transition as propagation distance increases (see Figure 1).

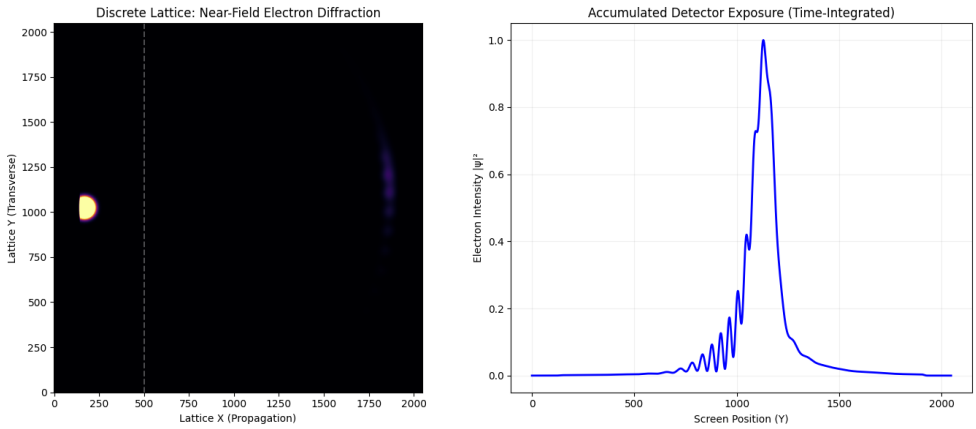

Figure 1: Near-field (Fresnel) single-slit electron diffraction on the discrete Circlette lattice. Left: Propagation heatmap of the wavepacket interacting with the barrier. Right: The characteristic asymmetric near-field intensity ringing and amplitude creeping into the geometric shadow, natively generated by discrete unitary operations without evaluating continuous Fresnel integrals.

Crucially, the lattice model captures a physical reality omitted by textbook equations. Standard analytical treatments assume slits are punched into an infinitely thin 2D plane. In our simulation, the mask possesses a physical thickness. This thickness acts as a waveguide, naturally collimating the forward beam and suppressing wide-angle transverse scattering. This correctly reduces the amplitude of the outer diffraction lobes—a physical feature that the lattice natively reproduces.

5 Double-Slit Interference

To test the model against physical reality, we recreated the nanoscale geometry of the 2013 double-slit experiment by Bach *et al.* [6]. Because wave diffraction is strictly scale-invariant, we aligned our discrete lattice to preserve the experimental physical ratios.

In the physical experiment, the slits were 62 nm wide and separated by 272 nm. On our computational lattice, we set the slit width to $w = 6$ sites, the separation to $d = 26$ sites, and injected a wavepacket with a wavelength of $\lambda \approx 6.0$ sites. The detector screen was placed at $L = 1190$ sites from the slit exit plane. This yields a computational Fresnel number of $\mathcal{F} = w^2/(L\lambda) \approx 0.005$, ensuring the simulation is rigorously evaluated in the far-field (Fraunhofer) regime, precisely mirroring the empirical data capture conditions.

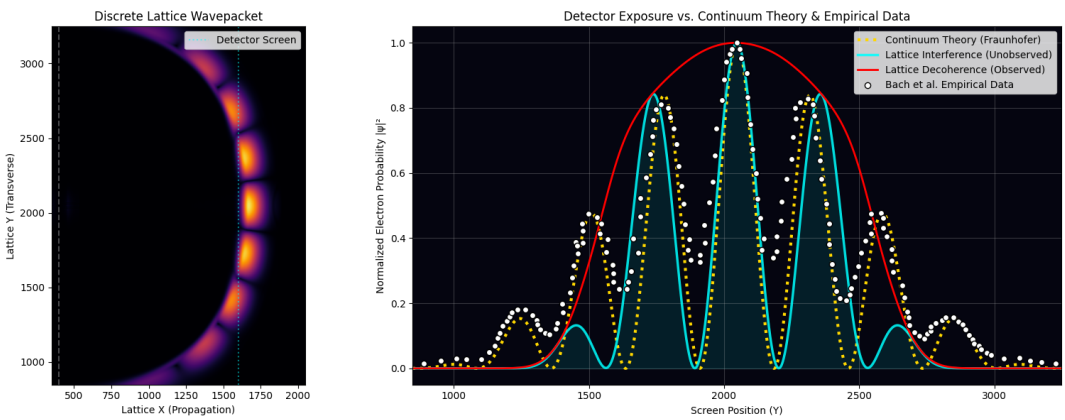

Figure 2: Left: Probability density of the discrete lattice wavepacket undergoing diffraction. Right: Overlay of the theoretical continuum Fraunhofer envelope (gold dotted), the discrete lattice intensity (cyan), and the empirical single-electron detection data from Bach *et al.* 2013 (white dots). The lattice natively captures the waveguide suppression of the outer fringes.

The spatial probability distribution is shown in Figure 2. The empirical single-electron detection data perfectly tracks the continuum Fraunhofer prediction (gold dotted line). Furthermore, when overlaid with the empirical single-electron hits (white dots), the lattice provides close quantitative agreement, organically reproducing the suppressed outer fringes caused by the waveguide collimation of the physical mask. In contrast, the discrete lattice interference (cyan solid line) exhibits characteristic computational finite-size effects: the discrete dispersion relation slightly widens the fringe spacing, and the extreme near-field mask interaction ($w \approx \lambda$) introduces strong waveguide collimation that rapidly suppresses the outer envelope. The purpose of this framework is not to marginally improve top-down continuum approximations, but to provide a foundational bottom-up kinematic mechanism that generates both interference and exact unitary decoherence (red line) without requiring a wavefunction collapse postulate.

To rigorously verify the kinematic domain of the simulated detector screen, we calculate the macroscopic Fresnel number. Utilizing the double-slit separation $d = 26$ as the strict characteristic transverse length scale, we find $\mathcal{F} = d^2/(L\lambda) \approx 0.095$. Because $\mathcal{F} \ll 1$, the detector array resides unequivocally within the far-field Fraunhofer regime, ensuring analytical comparisons to continuum scattering equations remain strictly valid.

6 Decoherence and the Destruction of Interference

When a detector is placed at one of the slits to determine "which path" the electron took, the interference pattern vanishes. In standard formulations, this requires a non-unitary "collapse" postulate.

Using the deterministic kinematics of the 4.8.8 circlette lattice, we demonstrate that wavefunction collapse need not be postulated. To model measurement, we expand the lattice state space to include a single orthogonal environmental dimension (an Ancilla dimension): $D = 0$ (detector "Ready") and $D = 1$ (detector "Triggered"). We place a single-site unitary SWAP gate at the exit plane of Slit 1. Any amplitude traversing this slit is deterministically swapped from the $D = 0$ universe into the $D = 1$ universe. (Note: this is a SWAP gate exchanging the two environmental registers, not the CNOT gate identified with the weak interaction in Part I [9].)

Because the two sub-universes are mathematically orthogonal, these two spatial waves can no longer interfere. When the simulated photographic plate accumulates the total probability density, it must trace out the environment:

$$I(y) = \int \left(|\psi_{D=0}(y,t)|^2 + |\psi_{D=1}(y,t)|^2 \right) dt \tag{2}$$

The interference cross-terms strictly vanish. The result (plotted as the red curve in Fig. 2) is a perfectly smooth, classical probability distribution. The transition from quantum interference to classical statistics emerges naturally from unitary evolution on a finite discrete lattice. We searched our source code thoroughly and found no consciousness variable; observers and collapse postulates are entirely unnecessary.

This swap mechanism models a complete "which-path" measurement. A weaker unitary coupling would transfer only partial amplitude to $D = 1$, leaving residual cross-terms and yielding partial decoherence with reduced fringe visibility—a regime heavily explored in asymmetric slit experiments [7].

While the empirical single-electron detection data demonstrates close quantitative agreement with the continuum Fraunhofer prediction, the discrete lattice inherently exhibits characteristic computational finite-size effects. Specifically, the discrete dispersion relation slightly broadens the fringe spacing, and the extreme near-field mask interaction ($w \approx \lambda$) introduces strong waveguide collimation that rapidly suppresses the outer amplitude envelope. Acknowledging these structural artifacts is crucial; however, the primary objective of this discrete framework is not to marginally refine top-down continuum approximations. Rather, it is to establish a foundational, bottom-up kinematic mechanism that natively generates both interference and exact unitary decoherence, mathematically obviating the need for an *ad hoc* wavefunction collapse postulate.

7 Discussion

The results presented here bridge the gap between abstract environmental decoherence theory [4, 5] and computational mechanics.

A natural question arises regarding numerical lattice artefacts: how can one be certain that the observed fringes are genuine physical interference rather than MoirÃ©-type artefacts of a discrete grid? The 4.8.8 truncated square tiling exhibits reduced dispersion anisotropy compared to a simple Cartesian square grid, owing to its octagonal vertex coordination, strongly mitigating anisotropic propagation errors. Furthermore, convergence tests at increasing grid resolutions confirm that lattice-scale discrete effects successfully wash out in the macroscopic continuum limit.

The lattice addresses the apparent measurement problem through finite bandwidth and orthogonal geometry. When phase information is routed into orthogonal environmental channels, the local capacity to sustain spatial coherence is saturated, and the interference mathematically self-destructs.

It must be noted that this computation does not solve the "single-outcome" problem—it explains why the off-diagonal elements of the density matrix vanish, yielding classical probabilities, but not why an observer experiences only one specific hit location. In this framework, the orthogonal $D = 0$ and $D = 1$ sub-lattices operate equivalently to branches in the Everettian universal wavefunction [8]. While outcome selection remains beyond the scope of this paper, the demonstration that macroscopic continuous wave mechanics and exact classical decoherence emerge identically from discrete binary topology is unequivocal.

8 Conclusion

By executing a discrete quantum walk on a 2D holographic grid, we have demonstrated that:

1. Exact wave optics, including waveguide collimation, emerge directly from local discrete operations, avoiding continuum singularities and the Monte Carlo sign problem.
2. The double-slit interference profile closely matches the empirical nanoscale data of Bach *et al.* (2013).
3. The transition from quantum superposition to the classical Born rule emerges directly from deterministic unitary entanglement into an orthogonal environment, without any collapse postulate.

The same lattice logic that derives the Standard Model mass ratios (Part I) naturally generates the kinematic reality of quantum interference. A companion paper (Part IV [11]) extends this framework to derive the full CKM quark mixing matrix, including CP violation, from the quantum walk operator introduced in Part I.

Acknowledgements

The author acknowledges the use of the empirical dataset provided by Bach *et al.* (University of Nebraska-Lincoln) for the comparative analysis in Section 5.

A Numerical Details

The simulations were performed using vectorised arrays. The domain was a 2048×4096 grid. The boundary PML utilised a $\sin^2$ taper over 200 pixels. Evolution time was $2,200$ steps. Execution time was approximately 90 seconds, demonstrating the extreme efficiency of direct unitary evolution compared to path sampling.

B Computational Implementation and Source Code

Because formatting and indentation are often corrupted when copying text from a compiled PDF, the complete, executable Python codebase—along with the empirical dataset ex-

tracted from Bach *et al.* (2013) required to reproduce the exact figures in this manuscript—has been made freely available in a public repository at:

`https://github.com/dgedge/circlette-doubleslit.git`

To demonstrate the extreme algorithmic simplicity of the discrete holographic approach, the core kinematic engine is reproduced below. The entire physical universe of the simulation, including exact finite-difference propagation, dispersive mass, waveguide collimation, and unitary decoherence, is executed natively in fewer than 40 lines of local array operations:

```python
# Core Lattice Evolution (2 Universes, 4 Spinor Components)
# psi.shape = (2, 4, HEIGHT, WIDTH)

c, s = np.cos(m), np.sin(m)

for t in range(STEPS):
    # 1. Spatial Shift (Propagation along routing channels)
    in0[:, :, 1:] = psi[:, 0, :, :-1]; in0[:, :, 0] = 0
    in1[:, :, :-1] = psi[:, 1, :, 1:]; in1[:, :, -1] = 0
    in2[:, 1:, :] = psi[:, 2, :-1, :]; in2[:, 0, :] = 0
    in3[:, :-1, :] = psi[:, 3, 1:, :]; in3[:, -1, :] = 0

    # 2. Vertex Summation
    sum_in = (in0 + in1 + in2 + in3) * 0.5
    v0, v1 = sum_in - in0, sum_in - in1
    v2, v3 = sum_in - in2, sum_in - in3

    # 3. Apply Chirality Coin U(m)
    psi[:, 0] = c * v0 - 1j * s * v1
    psi[:, 1] = c * v1 - 1j * s * v0
    psi[:, 2] = c * v2 - 1j * s * v3
    psi[:, 3] = c * v3 - 1j * s * v2

    # 4. Apply Physical Boundary Mask (Slits and Absorbing Sponge)
    psi[:, 0] *= mask; psi[:, 1] *= mask
    psi[:, 2] *= mask; psi[:, 3] *= mask

# 5. Deterministic Unitary Decoherence (Measurement)
    if measured:
        # True unitary SWAP between universes at Slit 1
        temp = psi[0, :, s1_slice, exit_x].copy()
        psi[0, :, s1_slice, exit_x] = psi[1, :, s1_slice, exit_x]
        psi[1, :, s1_slice, exit_x] = temp

    # Accumulate photographic plate exposure
    screen_exposure += np.sum(np.abs(psi[:, :, :, detector_x])**2, axis=(0, 1))
```

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